

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2269

Stateline 138 kV
November 03, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2269. The Interconnection Customer proposes a 260.1 MW solar photovoltaic facility interconnecting at the Stateline 138 kV substation in the Kentucky area. The request was received on December 29, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 782 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,330,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SPORN SUBSTATION 345 kV/230 kV XFMR
Contingency: COULEE AVE SUBSTATION-SPORN 345kV Line
Loading: 117.5% (Rating: 584 MVA)
DFAX: 5.3%
Pre-project loading: 115.1%
2. Facility: TRISTATE 345 kV/1 kV Autotransformer
Contingency: TRISTATE-HUBBARDSTOWN 345kV
Loading: 117.3% (Rating: 623 MVA)
DFAX: 24.9%
Pre-project loading: 106.8%
3. Facility: TRISTATE 138/1 kV Transformer Bank
Contingency: HUBBARDSTOWN-TRISTATE 345kV
Loading: 117.3% (Rating: 623 MVA)
DFAX: 24.9%
Pre-project loading: 106.8%
4. Facility: SUKUKA-EDUCATION SUBSTATION 138kV L
Contingency: VALEMOUNT-TRISTATE 345kV
Loading: 111.8% (Rating: 186 MVA)
DFAX: 7.4%
Pre-project loading: 101.4%
5. Facility: COMMERCIAL METALS-EDUCATION 138kV
Contingency: TRISTATE SUBSTATION-VALEMONT SUBSTATION 345kV L
Loading: 110.1% (Rating: 191 MVA)
DFAX: 7.4%
Pre-project loading: 100.0%

6. Facility: COMMERCIAL METALS-GABLE STREET 138kV
Contingency: VALEMONT-TRISTATE 345kV line
Loading: 108.5% (Rating: 196 MVA)
DFAX: 7.4%
Pre-project loading: 98.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SPORN SUBSTATION 345 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$13,510,000	8% (\$1,051,000)
TRISTATE 345 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$14,809,000	13% (\$1,861,000)
TRISTATE 138/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$5,348,000	13% (\$672,000)
SUKUKA-EDUCATION SUBSTATION 138kV L	Rebuild line with higher-capacity conductor	\$4,722,000	12% (\$587,000)
COMMERCIAL METALS-EDUCATION 138kV	Rebuild line with higher-capacity conductor	\$15,170,000	12% (\$1,884,000)
COMMERCIAL METALS-GABLE STREET 138kV	Reconductor line	\$2,211,000	12% (\$275,000)
	Total allocated to project		\$6,330,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.